MEC208 Lab 2: Control System CAD and CAS using MATLAB-Simulink

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***Abstract*—This report presents the modelling, stability assessment, controller design, and Simulink validation of an ID-specific DC motor drive supplied by an averaged bidirectional DC-DC converter. A second-order motor transfer function was derived and checked against the Simulink response. Proportional-gain tests showed that larger gains improve speed tracking but require larger transient armature current and stronger motor-voltage transients. A cascaded PI controller was then evaluated using the submitted Simulink plots.**

I. ID-Specific Lab Parameters

These unique parameters were generated using Ziwen Liao and Yijun Liu (my partner) XJTLU student IDs, 2361284 and 2361197, during the onsite lab on 2026-05-12: armature resistance $R_a = 7.7277\ \Omega$, armature inductance $L_a = 0.0265$ H, torque constant $K_t = 0.7571$ N m/A, back-EMF constant $K_e = 0.0793$ V/rpm, rotor inertia $J_{\text{rotor}} = 0.0153$ kg m^2, load coupled inertia $J_{\text{load}} = 0.0215$ kg m^2, system windage friction $B_{\text{sys}} = 7.0000{\times}10^{-4}$ N m s/rad, DC-DC buck converter switching inductor $L_{\text{conv}} = 3.8000{\times}10^{-6}$ H, input capacitance $C_1 = 4.1000{\times}10^{-6}$ F, output capacitance $C_2 = 4.8000{\times}10^{-6}$ F, DC bus input voltage $V_{\text{in}} = 250$ V, target settling time $T_s = 2.2000$ s, and target percent overshoot $M_p = 11.8000\%$.

The same MATLAB workspace also gives switching frequency $f_{\text{sw}} = 10000$ Hz, maximum motor current $I_{m,\max} = 7.5000$ A, rated current $I_{\text{rated}} = 2.5000$ A, rated speed $N_{\text{rated}} = 1750$ rpm, load disturbance = 2, current-loop gains $K_{p1} = 0.0200$, $K_{i1} = 3$, $K_{d1} = 0$, speed-loop gains $K_{p2} = 0.1000$, $K_{i2} = 0.0250$, $K_{d2} = 0$, time constant $T_{c1} = 0.4800$, control-loop sample time $T_{\text{sc}} = 1.0000{\times}10^{-4}$ s, Simulink computational time step $T_{\text{sim}} = 5.0000{\times}10^{-6}$ s, maximum simulation time $T_{\text{sim,max}} = 10$ s, and fixed DC bus value $V_{\text{in,fixed}} = 250$ V. The motor basic parameters further include rated power = 373 W, rated voltage = 180 V, and no-load speed = 1895 rpm. In the following modelling work, R_a is rounded to $7.728\ \Omega$ where the parameter generator prints three decimal places, and V_{in} and $V_{\text{in,fixed}}$ are treated as the same 250 V DC bus.

II. System Modelling with Simulink Validation

A. Physical System and Assumptions

The plant is a DC motor drive supplied by the averaged bidirectional DC-DC converter in the Simulink model. For modelling validation, the converter is represented by $V_m = V_{\text{in}}D$, the motor parameters are assumed constant, and viscous friction B_{sys} is used as the main mechanical loss.

B. Mathematical Model

The armature circuit is described by the applied motor voltage, the armature resistance and inductance, and the back-EMF generated by the motor speed:

$$V_m = L_a \frac{di_a}{dt} + R_a i_a + K_e \omega. \tag{1}$$

Here V_m, i_a, and ω denote motor voltage, armature current, and angular speed. Since K_e is given in V/rpm, it is converted as $K_e = 0.0793 \times 60/(2\pi) = 0.7573$ V/(rad/s). The mechanical equation is

$$J_{\text{eq}} \frac{d\omega}{dt} + B_{\text{sys}}\omega = K_t i_a, \quad J_{\text{eq}} = J_{\text{rotor}} + J_{\text{load}} = 0.0368 \text{ kg m}^2. \tag{2}$$

Taking the Laplace transform of (1)–(2) with zero initial conditions and eliminating $I_a(s)$ gives the speed plant:

$$\frac{\Omega_{\text{rpm}}(s)}{V_m(s)} = \frac{K_t(60/2\pi)}{L_a J_{\text{eq}} s^2 + (L_a B_{\text{sys}} + R_a J_{\text{eq}})s + R_a B_{\text{sys}} + K_t K_e}. \tag{3}$$

Substituting the generated parameters gives the speed and current models:

$$G(s) = \frac{\Omega_{\text{rpm}}(s)}{V_m(s)} = \frac{7.2298}{0.0009752 s^2 + 0.284409 s + 0.578731}. \tag{4}$$

Eliminating $\Omega(s)$ instead gives the armature-current model:

$$\frac{I_a(s)}{V_m(s)} = \frac{J_{\mathrm{eq}}s + B_{\mathrm{sys}}}{(L_a s + R_a)(J_{\mathrm{eq}}s + B_{\mathrm{sys}}) + K_t K_e} = \frac{0.0368s + 0.0007}{0.0009752s^2 + 0.284409s + 0.578731}. \tag{5}$$

$$\frac{I_a(s)}{D(s)} = \frac{9.20s + 0.175}{0.0009752s^2 + 0.284409s + 0.578731}. \tag{6}$$

Since $V_m = 250D$, the duty-to-speed plant is $G_d(s) = 250G(s)$:

$$\frac{\Omega_{\mathrm{rpm}}(s)}{D(s)} = \frac{1807.44}{0.0009752s^2 + 0.284409s + 0.578731}. \tag{7}$$

C. Simulink Validation

The Simulink model contains the physical DC Motor block and an equivalent signal-based validation path for the derived mathematical model using the same gains: $R_a = 7.728$, $1/L_a = 1/0.0265$, $K_t = 0.7571$, $1/J_{\mathrm{eq}} = 27.17$, $B_{\mathrm{sys}} = 0.0007$, $K_e = 0.7573$, and $60/(2\pi) = 9.5493$. Fig. 1, compares the physical model and derived model. The speed curves are almost superimposed, and the armature-current traces show the same initial peak and exponential decay, confirming both mechanical and electrical transient dynamics. The speed settles at approximately 3.1×10^3 rpm for a 250 V motor voltage, agreeing with the analytical DC gain $G(0) \times 250 = (7.2298/0.578731) \times 250 \approx 3123$ rpm.

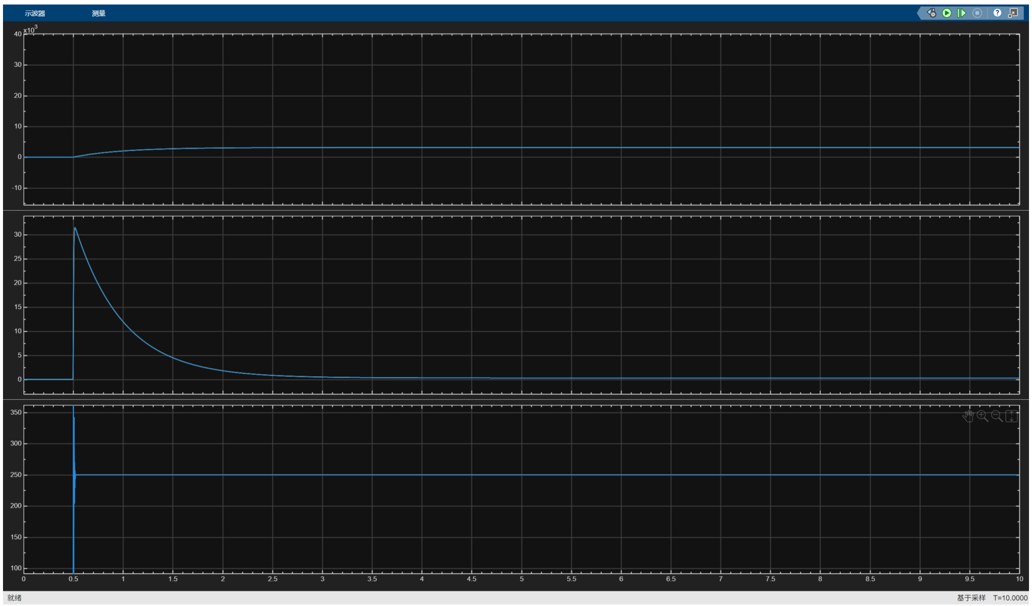


Fig. 1: System modelling validation comparing the Simulink physical model and the derived mathematical model responses. Blue traces represent the physical Simscape model, and yellow traces represent the derived mathematical model.

III. Sensitivity and Stability Analysis with Simulink Validation

A. Open-Loop and Closed-Loop Stability

The open-loop poles are obtained by setting the denominator of (4) to zero: $0.0009752s^2 + 0.284409s + 0.578731 = 0$. Solving this quadratic equation gives two negative real poles, approximately $s = -2.05$ and $s = -289.59$, so the voltage-to-speed plant is stable before feedback control is applied. However, open-loop operation cannot reject load disturbance or remove speed tracking error reliably, so closed-loop control is still required [1].

For the cascaded P/P structure used in Fig. 2, the speed controller output is the current reference and the current controller output is the duty command. Neglecting saturation, delay, and sampling, its approximate characteristic equation is

$$\begin{aligned} 0 = L_a J_{\mathrm{eq}} s^2 &+ \left[(R_a + V_{\mathrm{in}} K_{p1}) J_{\mathrm{eq}} + L_a B_{\mathrm{sys}}\right] s \\ &+ (R_a + V_{\mathrm{in}} K_{p1}) B_{\mathrm{sys}} + K_t K_e + K_t V_{\mathrm{in}} K_{p1} K_{p2}. \end{aligned} \tag{8}$$

Substituting the generated parameters gives

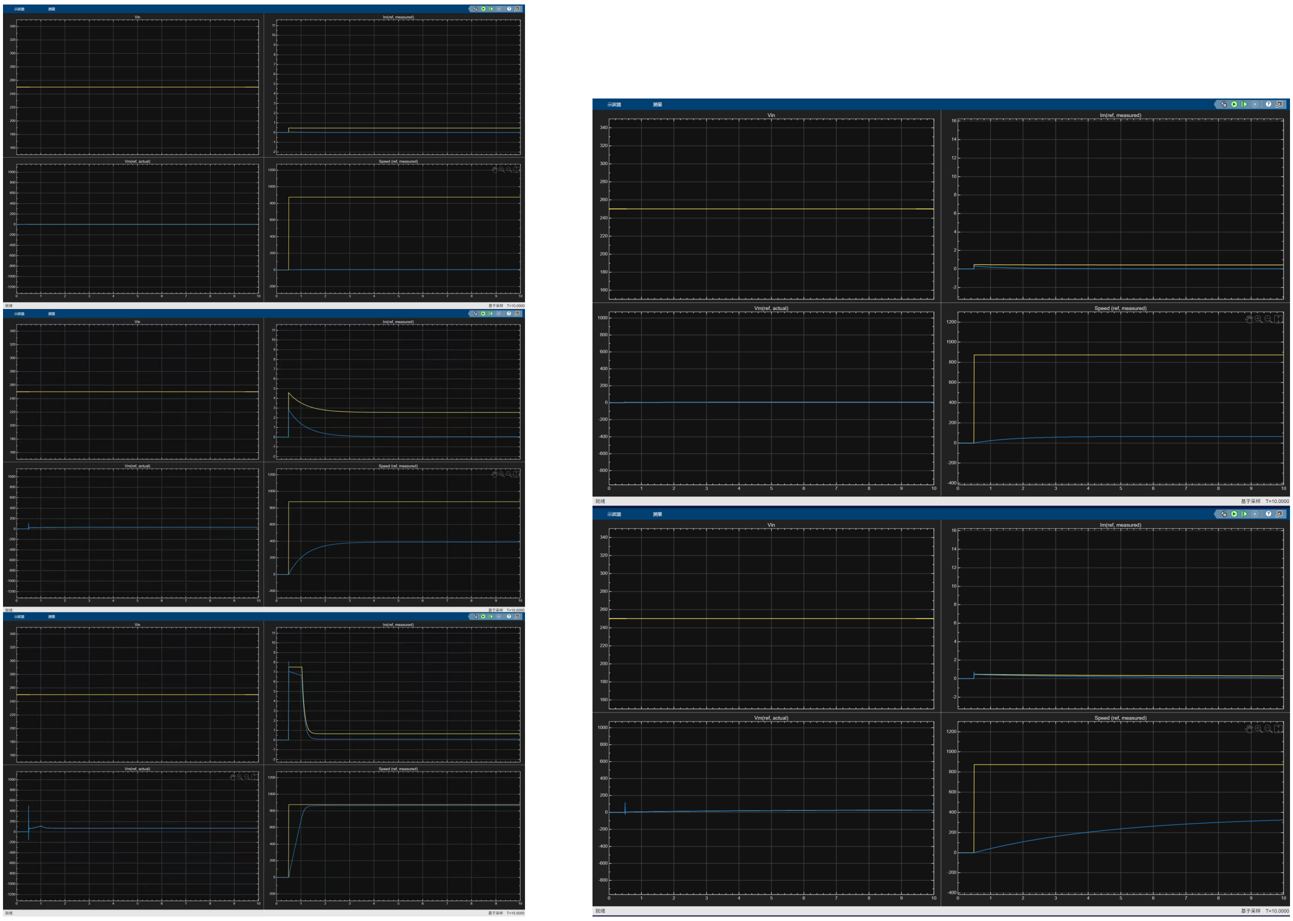
$$0 = 0.0009752s^2 + \left(0.284409 + 9.20K_{p1}\right) s + 0.578731 + 0.175K_{p1} + 189.28K_{p1}K_{p2}. \tag{9}$$

Equation (9) provides the theory link to the gain-sweep results. For positive K_{p1} and K_{p2}, all coefficients remain positive, so the nominal second-order approximation is stable. Increasing K_{p1} raises the coefficient of s and strengthens the inner current-loop damping effect, while increasing the product $K_{p1}K_{p2}$ raises the constant term and therefore the effective closed-loop stiffness. The theory therefore predicts that increasing both proportional gains should move the dominant response toward faster speed tracking, whereas increasing the current-loop gain alone cannot overcome a weak outer speed-loop command.

B. Sensitivity and Simulink Validation

The proportional-gain simulations in Fig. 2, verify these predicted trends with all I and D terms disabled. Fig. 2(a), increases the inner current-loop gain and outer speed-loop gain together, with $K_{p1} = K_{p2} = 0.005$, 0.05, and 0.5. As predicted by the $K_{p1}K_{p2}$ term in (9), the response changes from poor tracking at the smallest gains to much faster tracking at the largest gains. The cost is a larger electrical transient: a larger K_{p2} converts the speed error into a larger current reference, and a larger K_{p1} drives the duty command more aggressively to force the measured current toward that reference. Thus the improved speed response is obtained with greater armature-current and motor-voltage stress.

Fig. 2(b), keeps the outer speed-loop gain weak at $K_{p2} = 0.005$ and increases only the inner current-loop gain from $K_{p1} = 0.05$ to 0.5. The current loop reacts more strongly, but the speed response remains slow and far below the reference because the weak outer loop still generates only a small current reference. This agrees with (9), where speed-loop improvement depends strongly on the product $K_{p1}K_{p2}$. Although the nominal model remains stable for these positive gains, the trend should not be extrapolated indefinitely. Excessive gain raises closed-loop bandwidth and can reduce practical phase margin once converter delay, sampling, saturation, and unmodelled dynamics are included, so an overly aggressive setting may cause overshoot or oscillation even when the simplified continuous model appears stable. Besides controller gains, the response is also sensitive to plant parameters. A larger load inertia J_{load} would move the dominant mechanical pole closer to the imaginary axis and slow the speed response, while a higher armature resistance R_a would reduce available armature current for the same voltage command. Therefore, the gain trend observed in Fig. 2, should be interpreted together with possible plant-parameter variations in the real motor drive.



(a) $K_{p1} = K_{p2} = 0.005$, 0.05, and 0.5.

(b) $K_{p2} = 0.005$ with $K_{p1} = 0.05$ and 0.5.

Fig. 2: P-gain sensitivity tests for simultaneous loop-gain increase and inner-loop-only gain increase.

IV. Controller Selection, Gain Design, and Performance Evaluation with Simulink Validation

A. Controller Selection

The P-only tests in Fig. 2, show that speed tracking improves only when both loops are strengthened, but this increases current and motor-voltage transients and cannot remove steady-state speed error. A cascaded PI structure was therefore selected: the inner PI loop improves current tracking, while the outer PI loop removes the remaining speed error. Derivative action was omitted because it would amplify measurement noise and switching ripple.

B. Gain Design and PI Tuning

Guided by Fig. 2, the final gains were selected as a conservative compromise: larger outer-loop gains would command more current, while larger inner-loop gains would drive the duty command and motor voltage more strongly. The chosen gains therefore keep the current transient close to, but not far beyond, the practical `Im_max` = 7.5 A limit. The final workspace controller values are `Kp1` = 0.0200, `Ki1` = 3, `Kd1` = 0, `Kp2` = 0.1000, `Ki2` = 0.0250, and `Kd2` = 0. Since both derivative gains are zero, the controllers used for validation reduce to

$$D_1(s) = 0.02 + \frac{3.0}{s}, \quad D_2(s) = 0.10 + \frac{0.025}{s}. \tag{10}$$

Here $D_1(s)$ corresponds to the inner current loop and $D_2(s)$ corresponds to the outer speed loop in the Simulink model.

C. Simulink Performance Evaluation

The final PI/PI result is shown in Fig. 3. The design achieves small final speed error and an overshoot of about 5–6%, below `M_p` = 11.8%. The current response approaches the `Im_max` = 7.5 A limit, confirming that the gains are close to the electrical-stress constraint. The remaining weakness is settling time: the response is still outside the 2% band after `T_settling` = 2.2 s, so the design is a conservative current-limited trial rather than a fully optimized final controller. A faster version would need higher outer-loop gain with anti-windup and current-limit validation.

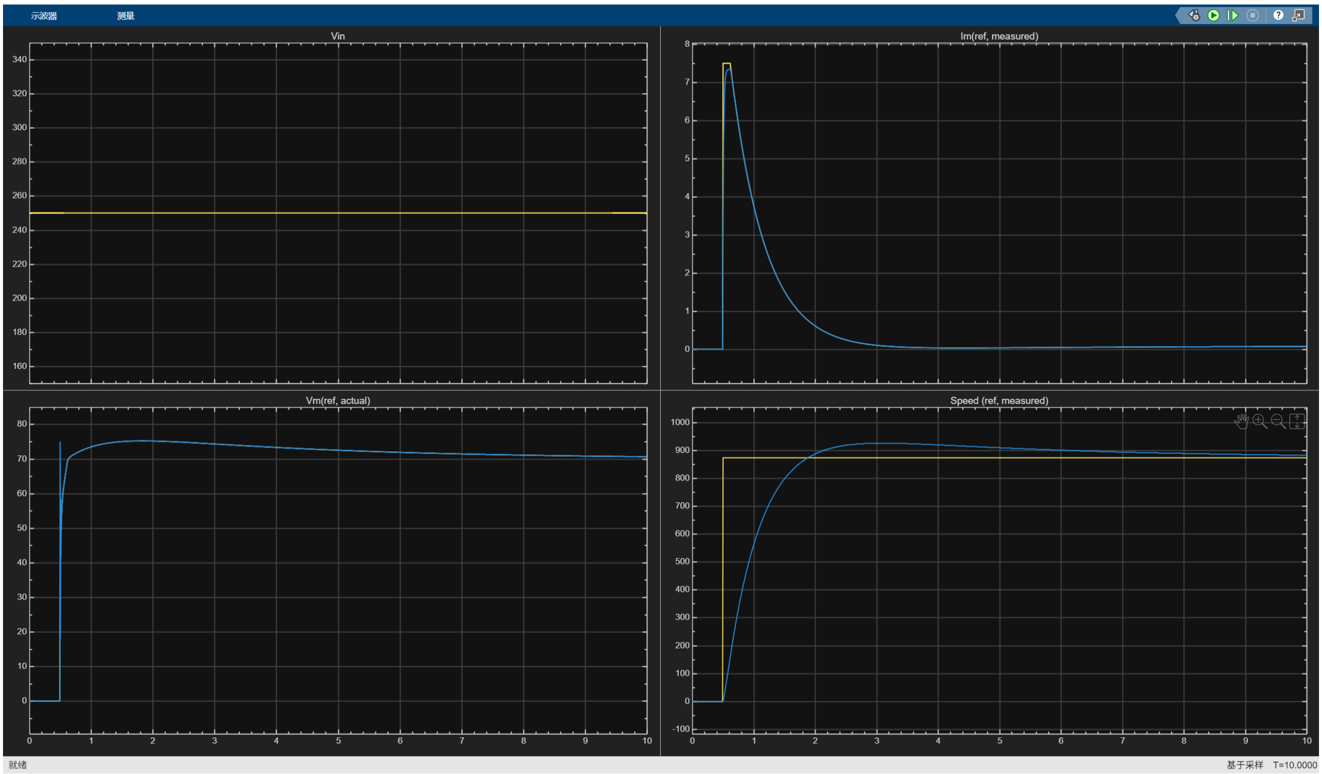


Fig. 3: Final cascaded PI control validation with `Kp1` = 0.0200, `Ki1` = 3, `Kd1` = 0, `Kp2` = 0.1000, `Ki2` = 0.0250, and `Kd2` = 0.

V. Discussion, Real-World Considerations

The nominal result in Fig. 3 must be judged against real-drive limits. The duty command is bounded by the 250 V DC bus and by converter current protection; if either limit is reached, the PI integrators may wind up and delay recovery unless anti-windup protection is added. The averaged converter model also omits switching ripple, dead time, voltage drops, sampling delay, quantization, and current/speed measurement noise, which can reduce phase margin and make aggressive gains less reliable than the ideal plots suggest. The fixed `Load_disturbance` = 2 applied at `Tsim_max`/2 would be a useful robustness test, because load torque changes, friction uncertainty, R_a heating, and variations in K_t, K_e, and J_{load} directly change current demand and settling. A better practical design would validate the controller under saturation, anti-windup, load disturbance, and parameter-perturbation tests, not only under the nominal step command.

Declaration

Generative AI was used only to learn relevant knowledge and basic MATLAB functions; all work is the author's.

References

[1] C. S. Lim, "MEC208 Lab 2: Control System CAD and CAS using MATLAB-Simulink," Xi'an Jiaotong-Liverpool University, 2025–2026.